Technical University of Denmark

Time Series Analysis 02417

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Assignment 3

ARMA, Seasonal Processes, and ARX Transfer Function Models

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1 Stability

We continue from Assignment 2 and consider the AR(2) process

$$X_t + \phi_1 X_{t-1} + \phi_2 X_{t-2} = \varepsilon_t, \qquad \varepsilon_t \sim \mathrm{WN}(0,1). \tag{1}$$

Equivalently,

$$X_t = -\phi_1 X_{t-1} - \phi_2 X_{t-2} + \varepsilon_t.$$

The theoretical autocorrelation function satisfies the Yule–Walker recursion

$$\rho(k) = -\phi_1 \rho(k-1) - \phi_2 \rho(k-2), \qquad k \geq 2, \tag{2}$$

with

$$\rho(0) = 1, \qquad \rho(1) = \frac{-\phi_1}{1+\phi_2}.$$

1.1 Simulate 5 realizations of the process

Using $n = 200$ observations, five realizations were simulated for $\phi_1 = -0.6$ and $\phi_2 = 0.5$. The roots of

$$1 + \phi_1 z + \phi_2 z^2 = 0$$

are complex with modulus $\sqrt{2} > 1$, so the process is stationary. The realizations fluctuate around zero and show stochastic oscillatory behavior but no explosive growth.

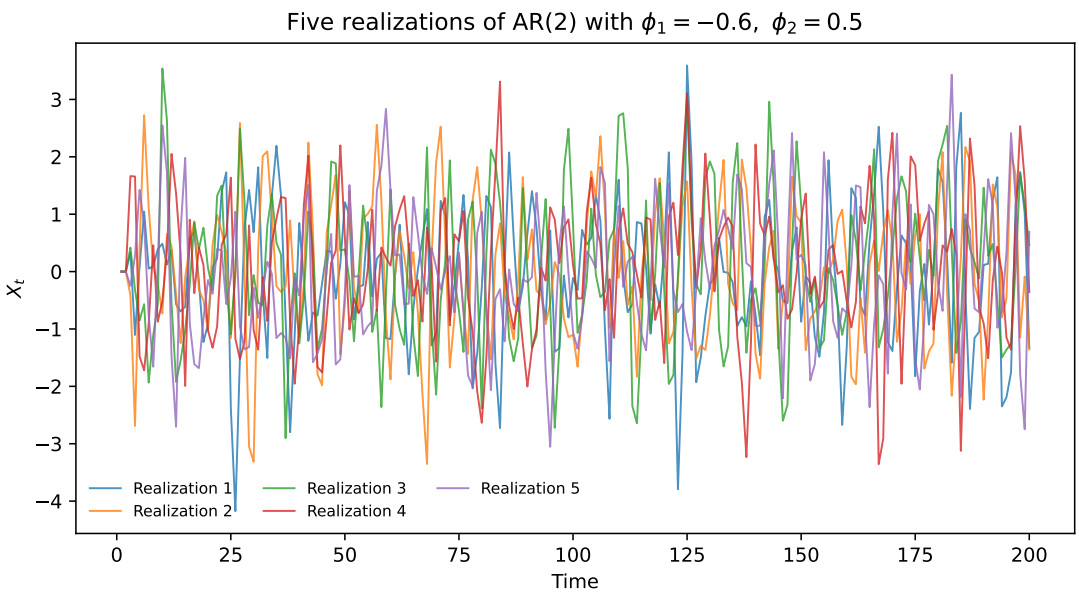

Figure 1: Five simulated realizations of the AR(2) process with $\phi_1 = -0.6$ and $\phi_2 = 0.5$.

1.2 Calculate the empirical ACF of the simulations and plot them together with $\rho(k)$

Figure 2 compares the empirical ACFs of the five realizations with the theoretical $\rho(k)$ up to lag 30. The empirical curves fluctuate around the theoretical ACF because the sample size is finite, but the overall damped oscillatory pattern is clearly reproduced. This is exactly what is expected for a stationary AR(2) process with complex roots.

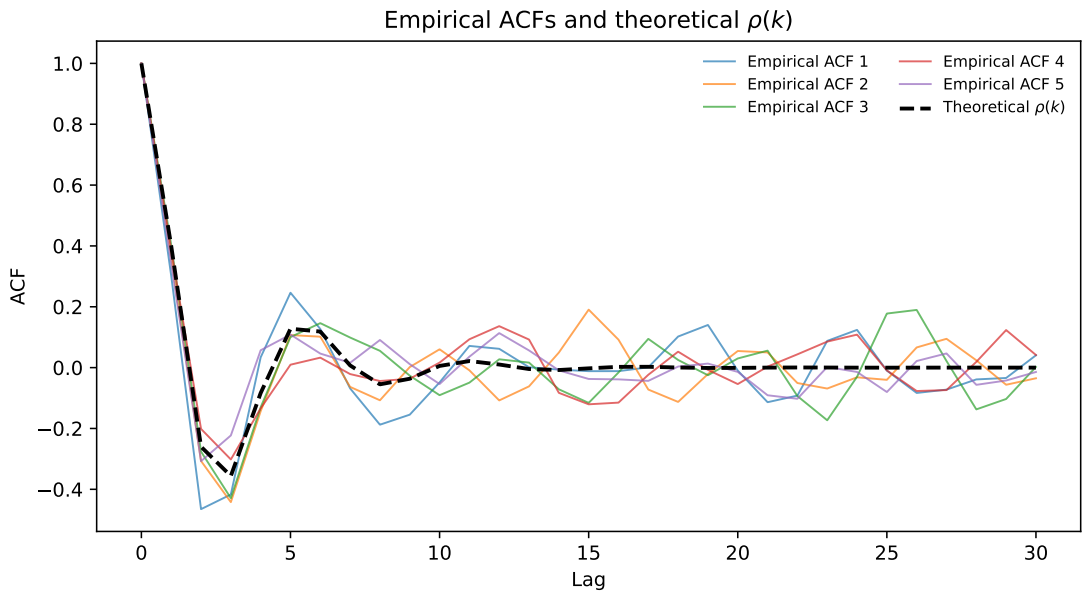

Figure 2: Empirical ACFs of the five realizations together with the theoretical $\rho(k)$ for $\phi_1 = -0.6$ and $\phi_2 = 0.5$.

1.3 Repeat the analysis for new coefficient values

The same simulation and ACF analysis was repeated for the four new coefficient settings.

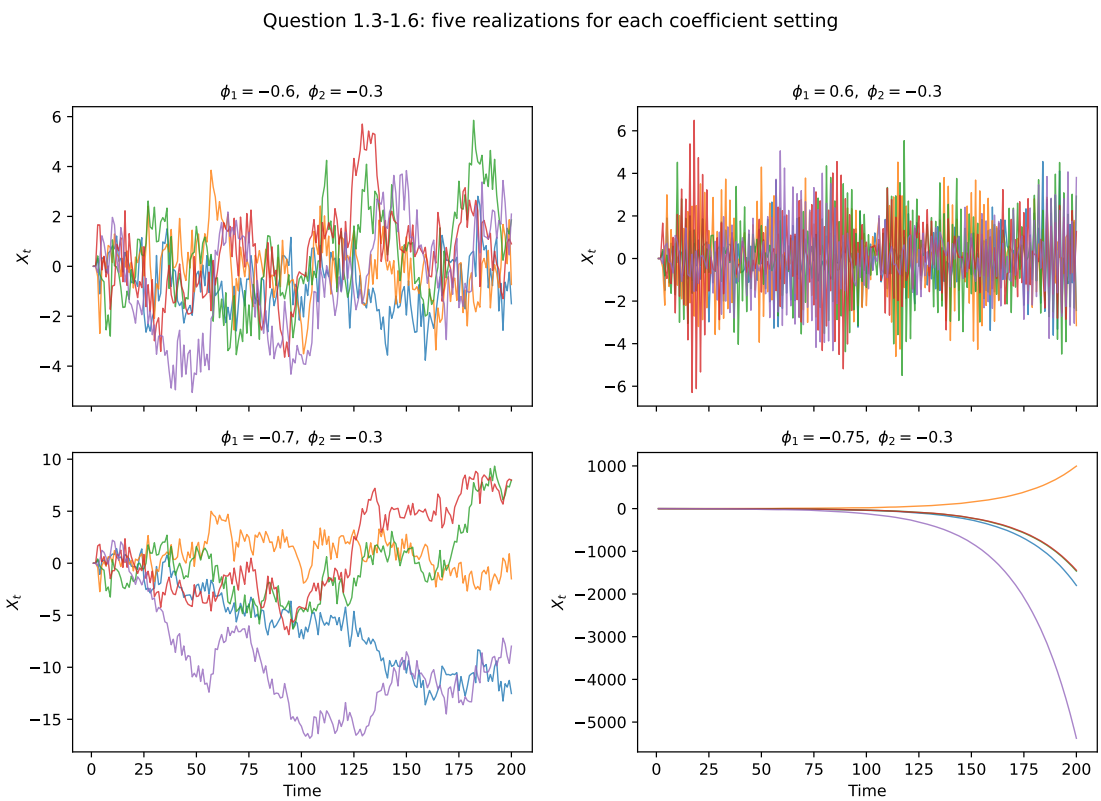

Figure 3: Five realizations for the four coefficient settings in Questions 1.3–1.6.

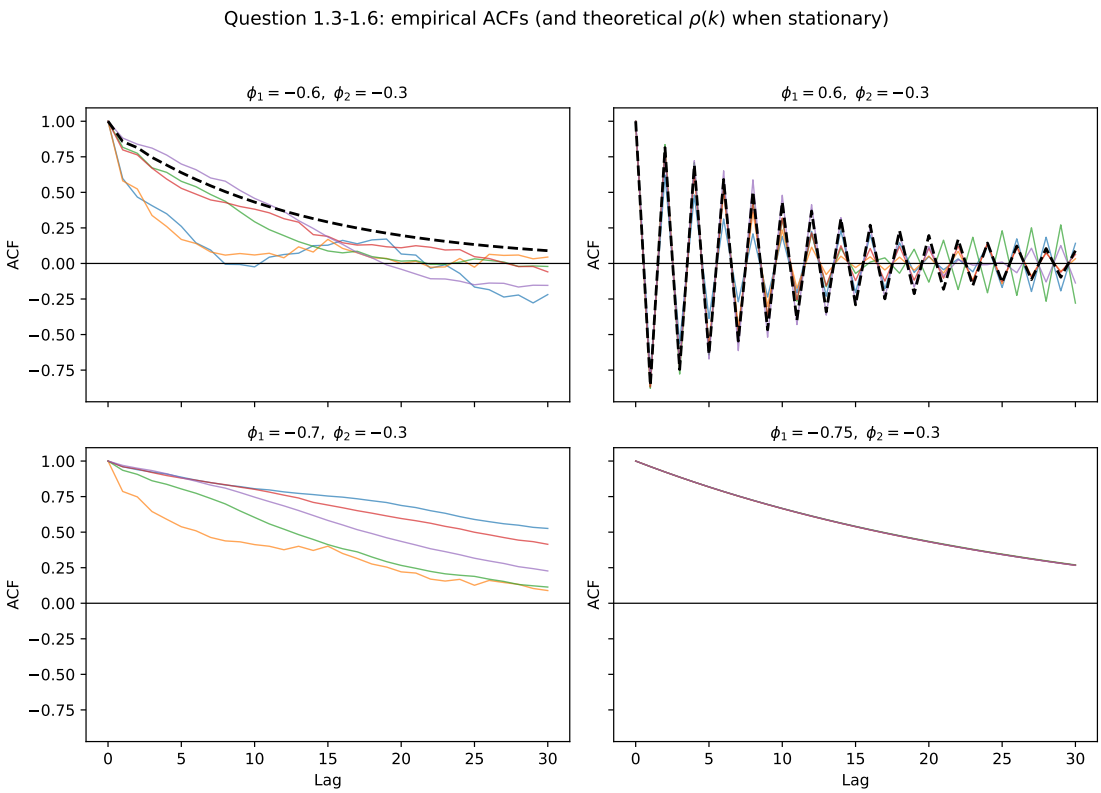

Figure 4: Empirical ACFs for Questions 1.3–1.6. The theoretical ACF is overlaid with a dashed line when the process is stationary.

1.3: $\phi_1 = -0.6$**,** $\phi_2 = -0.3$**.** This case is stationary because the two roots have modulus larger than 1. The realizations look stable, and the ACF is positive and decays smoothly toward zero. The empirical ACFs are close to the theoretical curve.

1.4: $\phi_1 = 0.6$**,** $\phi_2 = -0.3$**.** This process is also stationary, but now the signs of the coefficients generate an alternating dependence structure. The time series remains bounded, while the ACF oscillates in sign and decays gradually, which is characteristic of a stationary AR(2) process with opposite-sign persistence.

1.5: $\phi_1 = -0.7$**,** $\phi_2 = -0.3$**.** Here one root is exactly at 1, so the process is on the stationarity boundary. In the simulations the series behaves as a nearly non-stationary process, and the empirical ACF remains very large over many lags instead of decaying quickly. The theoretical stationary ACF is therefore no longer a reliable reference.

1.6: $\phi_1 = -0.75$**,** $\phi_2 = -0.3$**.** This case is non-stationary because one root lies inside the unit circle. The simulated realizations show explosive behavior, and the empirical ACF stays close to 1 for many lags. This confirms that the process is not stable.

1.4 Should the time series plot always be examined, or is the ACF sufficient?

The ACF is essential for identifying dependence structure, but it is not sufficient on its own. The time-series plot reveals behaviors such as explosive growth, level changes, or other obvious non-stationary features that may be hard to interpret from the ACF alone. In this exercise the distinction between the boundary case in 1.5 and the explosive case in 1.6 is much easier to

understand when both the time series and the ACF are examined together. Therefore, I would recommend always looking at both.

2 Predicting monthly solar power

The identified model is

$$(1+\phi_1 B)(1+\Phi_1 B^{12})(\log(Y_t)-\mu)=\varepsilon_t, \tag{3}$$

with known parameters

$$\phi_1=-0.38, \qquad \Phi_1=-0.94, \qquad \mu=5.72,$$

and innovation variance $\sigma_\varepsilon^2=0.222$. Define

$$X_t=\log(Y_t)-\mu.$$

Then the model can be written as

$$X_t+\phi_1 X_{t-1}+\Phi_1 X_{t-12}+\phi_1\Phi_1 X_{t-13}=\varepsilon_t. \tag{4}$$

2.1 Re-write the model to calculate residuals and validate the model

From (4), the one-step residuals are computed as

$$\hat{\varepsilon}_{t|t-1}=X_t+\phi_1 X_{t-1}+\Phi_1 X_{t-12}+\phi_1\Phi_1 X_{t-13}, \qquad t=14,\ldots,36. \tag{5}$$

The resulting residuals are centered close to zero, with sample mean 0.013 and sample variance 0.044. A Ljung–Box test at lag 10 gives $p=0.250$, so there is no significant evidence of remaining serial correlation. The ACF is weak, and the Q–Q plot is acceptable given the very short residual series (23 effective residuals). Overall, the i.i.d. assumption is reasonably supported in terms of dependence, although the variance estimate is clearly more uncertain than the mean because of the short sample.

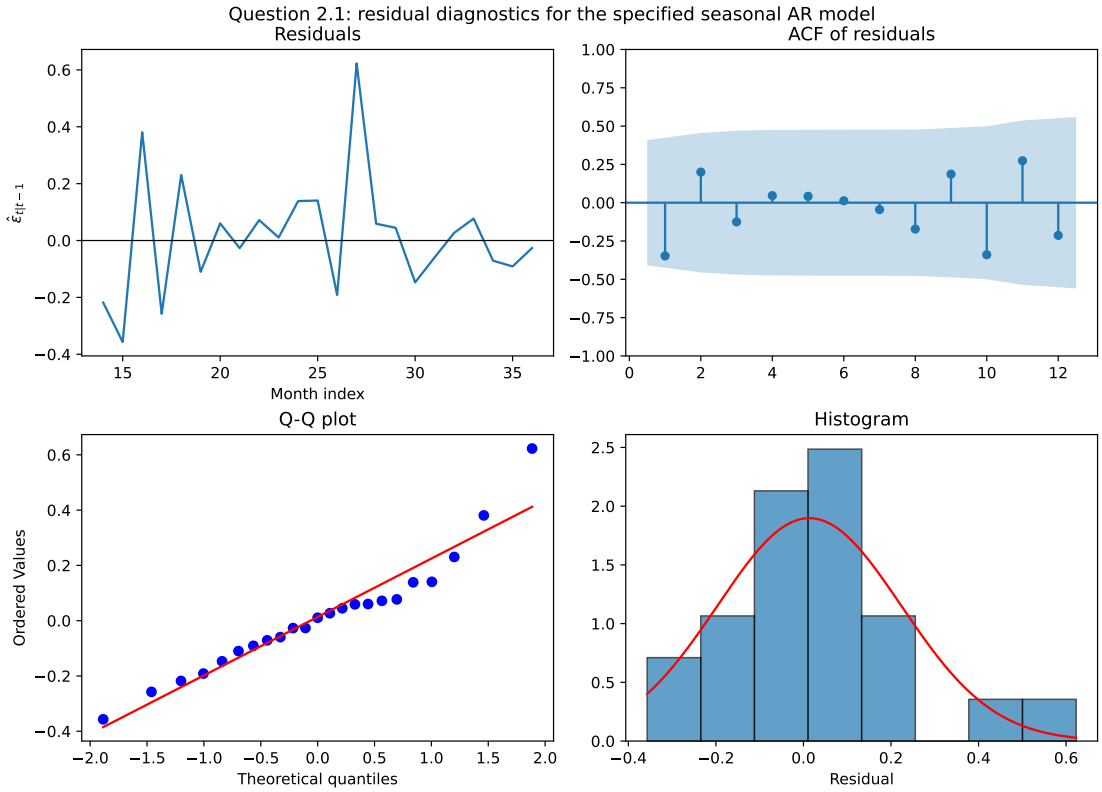

Figure 5: Residual diagnostics for the specified seasonal AR model: residual series, ACF, Q–Q plot, and histogram.

2.2 Calculate $\hat{Y}_{t+k|t}$ for $t = 36$ and $k = 1, \ldots, 12$

Because the forecast horizon is only 12 months, all seasonal lags needed in the recursion are observed values from the historical sample. Therefore the point forecasts on the X_t scale are obtained recursively from

$$\hat{X}_{36+k|36} = -\phi_1 \hat{X}_{35+k|36} - \Phi_1 X_{24+k} - \phi_1 \Phi_1 X_{23+k}, \qquad k = 1, \ldots, 12, \tag{6}$$

with back-transformation

$$\hat{Y}_{36+k|36} = \exp(\hat{X}_{36+k|36} + \mu). \tag{7}$$

The resulting forecasts are listed in Table 1. They reproduce the strong annual seasonal pattern of the plant, with low winter production and a summer peak.

Table 1: Twelve-month-ahead forecasts and 95% prediction intervals for monthly solar power.

Month	Forecast$_M Wh$	Lower$_9$5	Upper$_9$5
2011-01	101.28	40.22	255.02
2011-02	147.23	54.82	395.39
2011-03	156.92	57.91	425.22
2011-04	142.55	52.54	386.77
2011-05	355.31	130.93	964.22
2011-06	538.02	198.25	1460.09
2011-07	611.21	225.22	1658.71
2011-08	631.89	232.84	1714.86
2011-09	630.11	232.18	1710.02
2011-10	508.16	187.25	1379.06
2011-11	375.01	138.19	1017.72
2011-12	280.49	103.36	761.21

2.3 Calculate 95% prediction intervals and add them to the plot

Following the assignment instructions, the prediction interval is computed using only the AR(1) part

$$(1 + \phi_1 B)X_t = \varepsilon_t.$$

Hence the k-step forecast error variance is approximated by

$$\sigma_k^2 = \sigma_\varepsilon^2 \sum_{i=0}^{k-1} \phi_1^{2i}, \qquad k = 1, \ldots, 12. \tag{8}$$

A 95% interval on the log scale is then

$$\hat{X}_{36+k|36} \pm 1.96\, \sigma_k,$$

and the interval on the original scale is obtained by exponentiation.

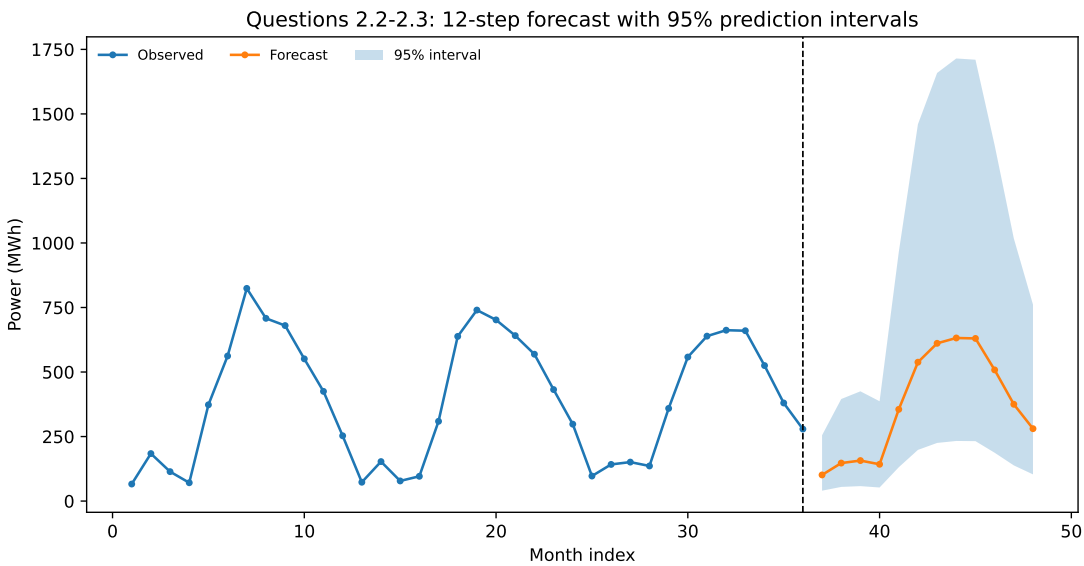

Figure 6: Observed monthly solar power together with the twelve-month forecast and 95% prediction intervals.

2.4 Comment on the forecast and the interval width

The point forecast is plausible because it follows the seasonal structure in the historical series and the residual diagnostics do not indicate major remaining serial dependence. However, the uncertainty bands should be interpreted with care. First, the prediction intervals were computed from the AR(1) part only, as instructed, so they ignore additional uncertainty from the seasonal term. Second, parameter uncertainty is not included. Third, because the intervals are transformed back with the exponential function, they become highly asymmetric on the original scale and widen strongly when the forecast level is high. For these reasons, the intervals are useful as rough guidance but probably somewhat too optimistic in some periods and too wide in others on the original MWh scale.

3 An ARX model for the heating of a box

The goal is to predict the hourly heating demand Ph_t using the exogenous variables $Tdelta_t$ and Gv_t together with lagged values of the output. The last training observation is fixed at "2013-02-06 00:00", giving 167 training observations and 64 test observations.

3.1 Read the data and plot the three non-lagged time series

The three non-lagged time series are shown in Figure 7. The heating power Ph shows clear daily structure with repeated dips and peaks. $Tdelta$ is persistent and changes much more slowly. Gv is zero or close to zero during the night and peaks during daylight hours. Visually, higher solar radiation coincides with reduced heating demand, which already suggests a negative effect of Gv on Ph.

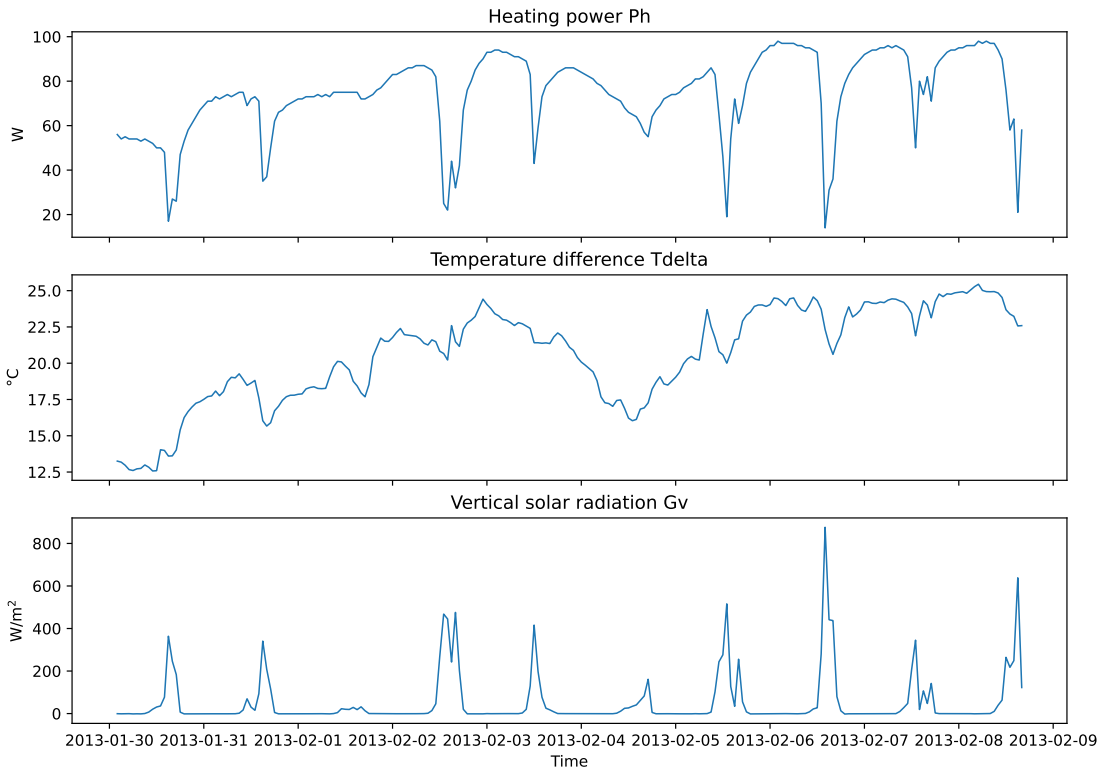

Figure 7: The three observed non-lagged time series: heating power Ph, temperature difference $Tdelta$, and vertical solar radiation Gv.

3.2 Split into training and test sets

The data was split exactly as requested:

$$\text{training set: } t = 1, \ldots, 167, \qquad \text{test set: } t = 168, \ldots, 231.$$

All model estimation and diagnostic work below is performed on the training set unless explicitly stated otherwise.

3.3 Investigate the variables and their relations

Figure 8 summarizes the most important exploratory plots. The scatter plots show a positive relation between Ph and $Tdelta$ and a negative relation between Ph and Gv. The ACF of Ph shows strong serial dependence and a clear daily pattern. The ACF of $Tdelta$ decays slowly, confirming that temperature difference is highly persistent. The CCF between $Tdelta$ and Ph is strongest around lag 0 and nearby lags, while the CCF between Gv and Ph is strongly negative near lag 0. These findings strongly support a transfer-function / ARX formulation rather than a static regression only.

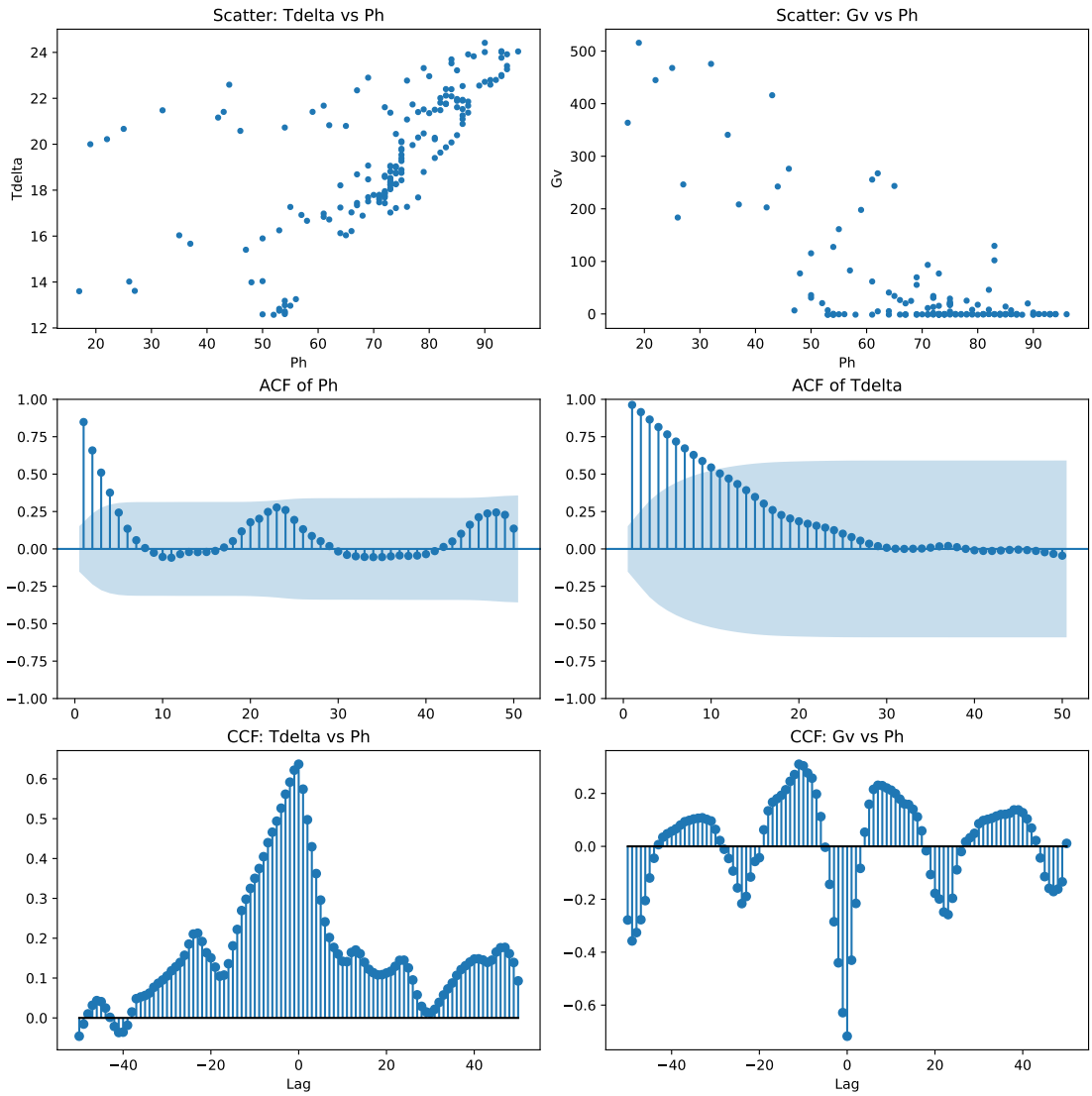

Figure 8: Scatter plots, ACFs, and CCFs for the main variables in the training set.

3.4 Estimate the impulse response from $Tdelta$ and Gv to Ph up to lag 10

To obtain simple empirical impulse-response estimates, finite distributed-lag regressions were fitted using lags 0 to 10 of each input separately. The resulting coefficients are shown in Figure 9. The estimated response from $Tdelta$ to Ph is positive and distributed over several hours, with the largest weights at lags 0, 1, and 7. In contrast, the effect of Gv is dominated by a contemporaneous negative coefficient at lag 0 and much smaller lagged effects thereafter. This is physically plausible: stronger solar radiation reduces the heating need immediately, while changes in temperature difference have a broader delayed effect.

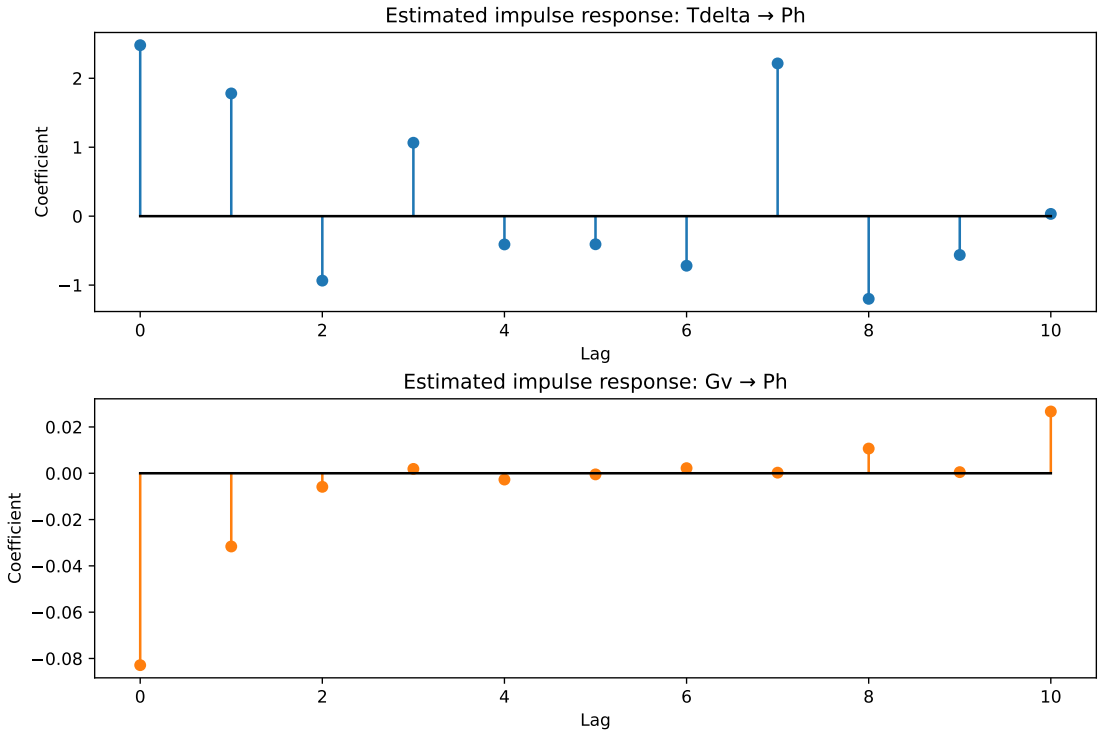

Figure 9: Estimated finite impulse responses from $Tdelta$ and Gv to Ph up to lag 10.

3.5 Fit the linear regression model

The static linear regression model is

$$Ph_t = \omega_1 Tdelta_t + \omega_2 Gv_t + \varepsilon_t, \tag{9}$$

with estimated coefficients

$$\hat{\omega}_0 = 11.148, \qquad \hat{\omega}_1 = 3.336, \qquad \hat{\omega}_2 = -0.111.$$

The fitted values capture the broad level of the heating demand, but the one-step residuals still show strong autocorrelation. The expanding-window one-step RMSE is 5.465, and a Ljung–Box test at lag 24 gives a p-value below 10^{-16}. The CCFs between the residuals and the inputs also remain clearly non-zero. Hence the static regression does not explain the dynamic dependence structure well enough, and a model with output feedback is needed.

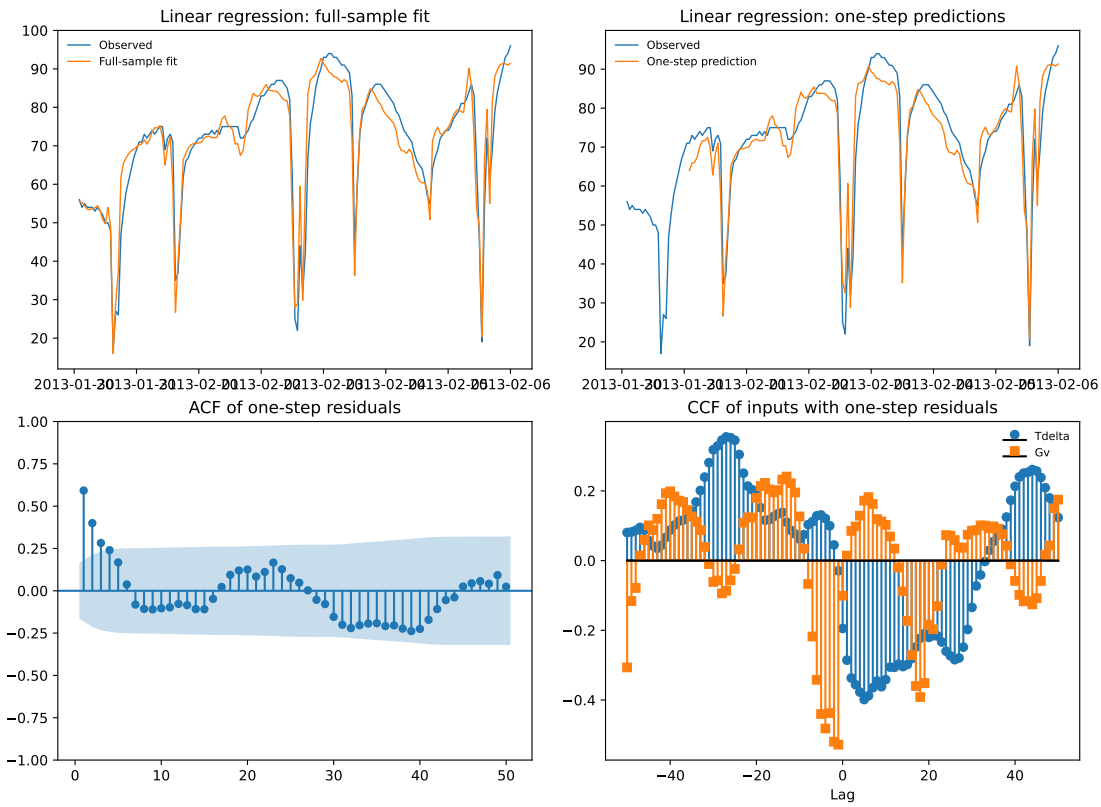

Figure 10: Linear regression diagnostics: full-sample fit, one-step predictions, residual ACF, and residual CCFs.

3.6 Fit the first-order ARX model

The first-order ARX model is

$$Ph_t = -\phi_1 Ph_{t-1} + \omega_1 Tdelta_t + \omega_2 Gv_t + \varepsilon_t. \tag{10}$$

In the regression implementation the lagged output enters directly, so the fitted model is reported in terms of its estimated coefficient on Ph_{t-1}:

$$\hat{Ph}_t = 5.365 + 0.405\, Ph_{t-1} + 2.096\, Tdelta_t - 0.085\, Gv_t.$$

This model is a clear improvement. The one-step RMSE drops to 2.762, roughly half the value of the static regression. The residual standard deviation is much lower, and the Ljung–Box p-value at lag 24 is 0.550, which is consistent with white-noise residuals. The residual CCFs are also much smaller. Therefore, including one lag of the output captures most of the missing dynamics.

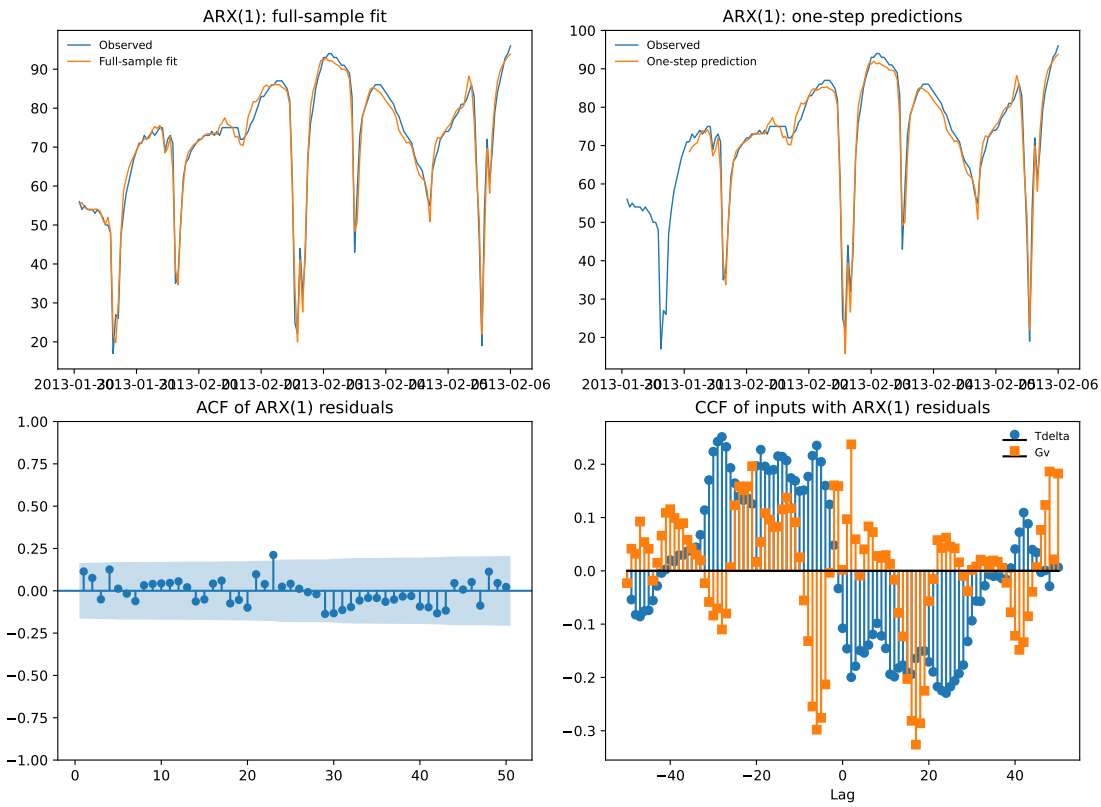

Figure 11: ARX(1) diagnostics: full-sample fit, one-step predictions, residual ACF, and residual CCFs.

3.7 Increase the ARX model order and compare AIC/BIC

For order r, the model class is

$$Ph_t = \sum_{j=1}^{r} a_j Ph_{t-j} + \sum_{j=0}^{r-1} b_j Tdelta_{t-j} + \sum_{j=0}^{r-1} c_j Gv_{t-j} + \varepsilon_t. \tag{11}$$

AIC and BIC were evaluated for orders 1 to 10. Figure 12 shows that both criteria decrease substantially at low orders and achieve their minima at order 6. As expected, BIC lies above AIC because it penalizes model complexity more strongly. In this dataset both criteria nevertheless select the same order, $r = 6$.

3.8 One-step predictions in the test period and RMSE versus model order

The right panel of Figure 12 shows the one-step test RMSE for orders 1 to 10. The minimum occurs at order 3, not at order 6. Hence the out-of-sample performance suggests a smaller model than the in-sample information criteria. The interpretation is standard: the larger order improves fit on the training sample, but part of that improvement does not transfer to the unseen test period. Therefore, I choose order 3 for the final simulation study.

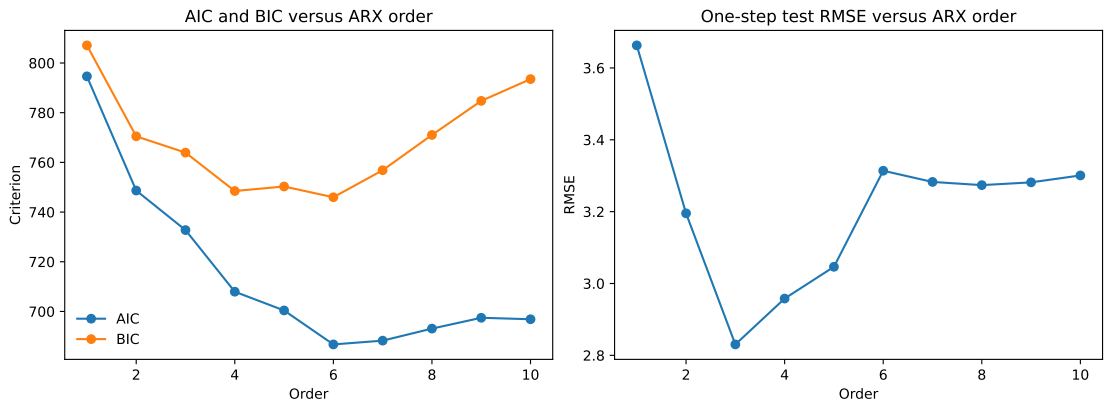

Figure 12: Left: AIC and BIC versus ARX order. Right: one-step test RMSE versus ARX order.

Table 2: Orders selected by the three model-selection criteria.

Criterion	Selected order	Value
AIC	6	686.727
BIC	6	745.969
Test RMSE	3	2.830

3.9 Multi-step prediction (simulation) through the full period

Using the order selected by test RMSE, the final model is ARX(3). Recursive simulation was performed over the full period using the observed inputs $Tdelta$ and Gv, while the lagged Ph terms were generated iteratively from previous simulated values. The fitted ARX(3) model is approximately

$$\begin{aligned}\hat{Ph}_t = {} & 3.634 + 0.321\,Ph_{t-1} + 0.150\,Ph_{t-2} + 0.011\,Ph_{t-3} \\ & + 0.069\,Tdelta_t + 1.447\,Tdelta_{t-1} + 0.374\,Tdelta_{t-2} \\ & - 0.098\,Gv_t + 0.000\,Gv_{t-1} + 0.020\,Gv_{t-2}.\end{aligned}$$

Figure 13 shows that the recursive simulation tracks the broad daily pattern well, but some sharp peaks and troughs are smoothed out. The full-path RMSE is 2.499 (excluding the 3 warm-up observations), and the RMSE on the test part of this recursive simulation is 3.354. The model is therefore useful for multi-step prediction, but the loss of accuracy relative to one-step prediction is clearly visible.

In a real-time operational setting, such multi-step predictions are only feasible if forecasts of the exogenous inputs are also available. In this exercise the observed future $Tdelta$ and Gv were used, which is appropriate for model evaluation but not directly available in real operation.

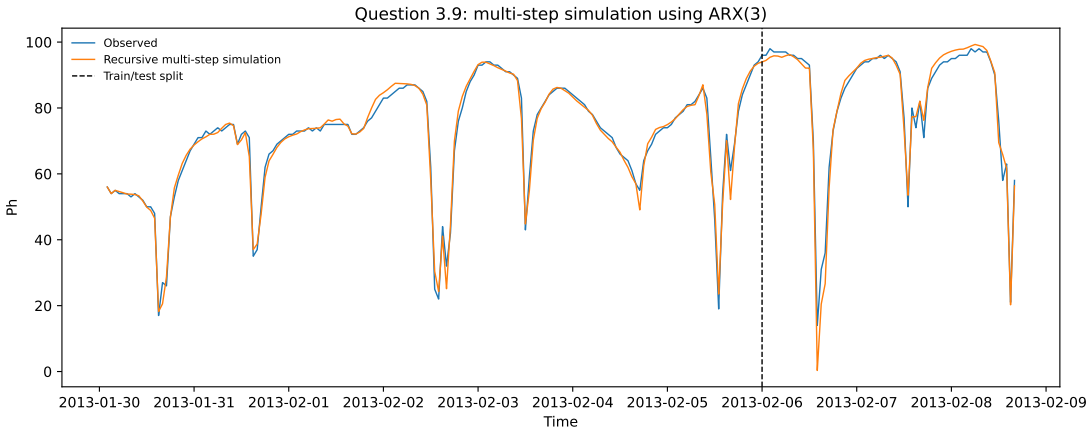

Figure 13: Recursive multi-step simulation using the selected ARX(3) model.

3.10 Summarizing conclusion

The heating data clearly requires a dynamic model. The static regression captures only the broad level and leaves strong structure in the residuals. Adding one lag of the output already improves the fit substantially and produces near-white residuals. When the model order is increased further, AIC and BIC prefer order 6, while out-of-sample one-step RMSE prefers order 3. I therefore consider ARX(3) the most suitable compromise in this assignment: it is small, predicts well on the test set, and gives a reasonable recursive multi-step simulation. Overall, the exercise illustrates the main course message clearly: in-sample fit, one-step prediction, and multi-step simulation answer different questions and may point to different "best" models.

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